

NEURO ANATOMY Volume-4

Endocrine System also transmit signal but it is not as fast as Nervous System.

If we have to take fast action like our hand is on the hot tawa then our spinal cord will response but when we have to make decisions, plans the over BRAIN involves.

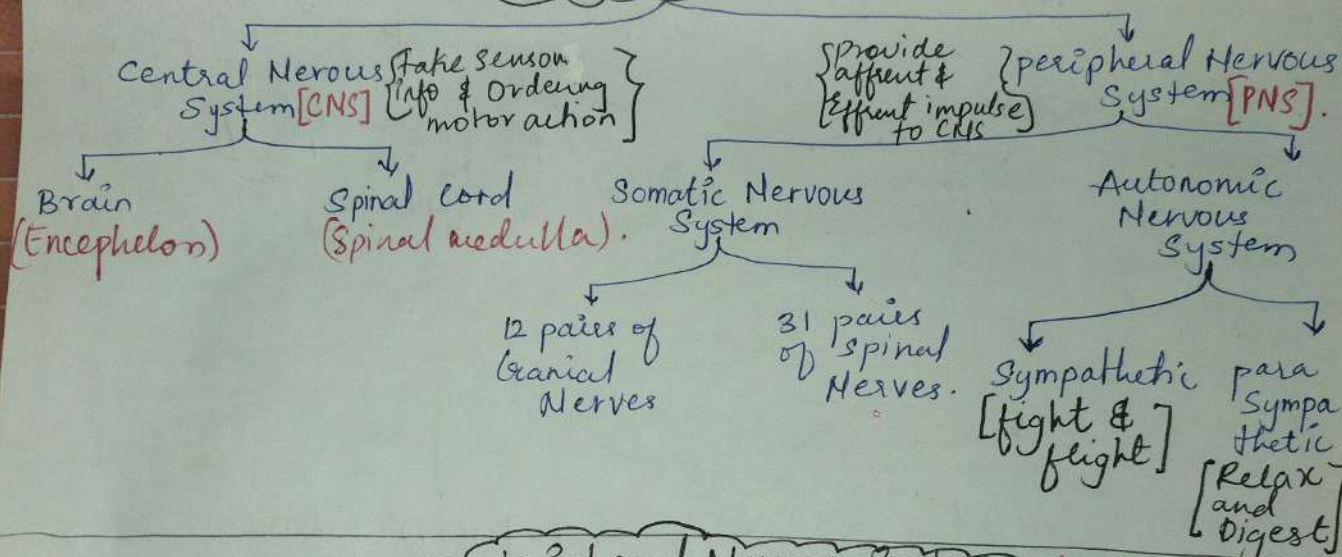
Nervous System

Nervous System is complex mechanism involves, Nerves, Neurons, Brain, Spinal Cord & Nerves that extend to the whole body

Example of switch on the button for light Nervous system take less time than switch on the button of light.

2nd example tawa → the way sensory organ (Skin) react when we touch the hot tawa. * receptors are present in all 5 Sensory Organs

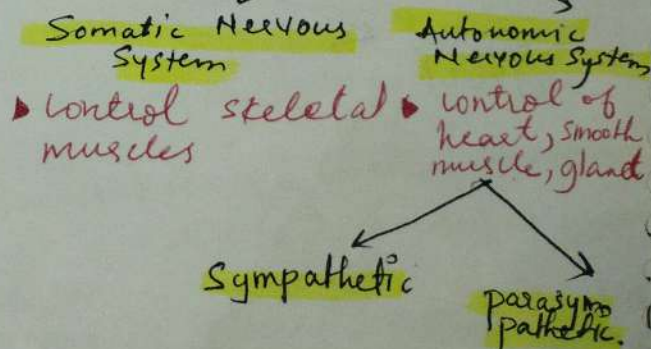
Nervous System (Anatomical division)



Peripheral Nervous System (Functional Division)

1) Afferent → provide sensory info to CNS.

2) Efferent → carries motor info to muscle, glands, blood vessel & heart via



Nervous Tissue

Neuron or nerve cells

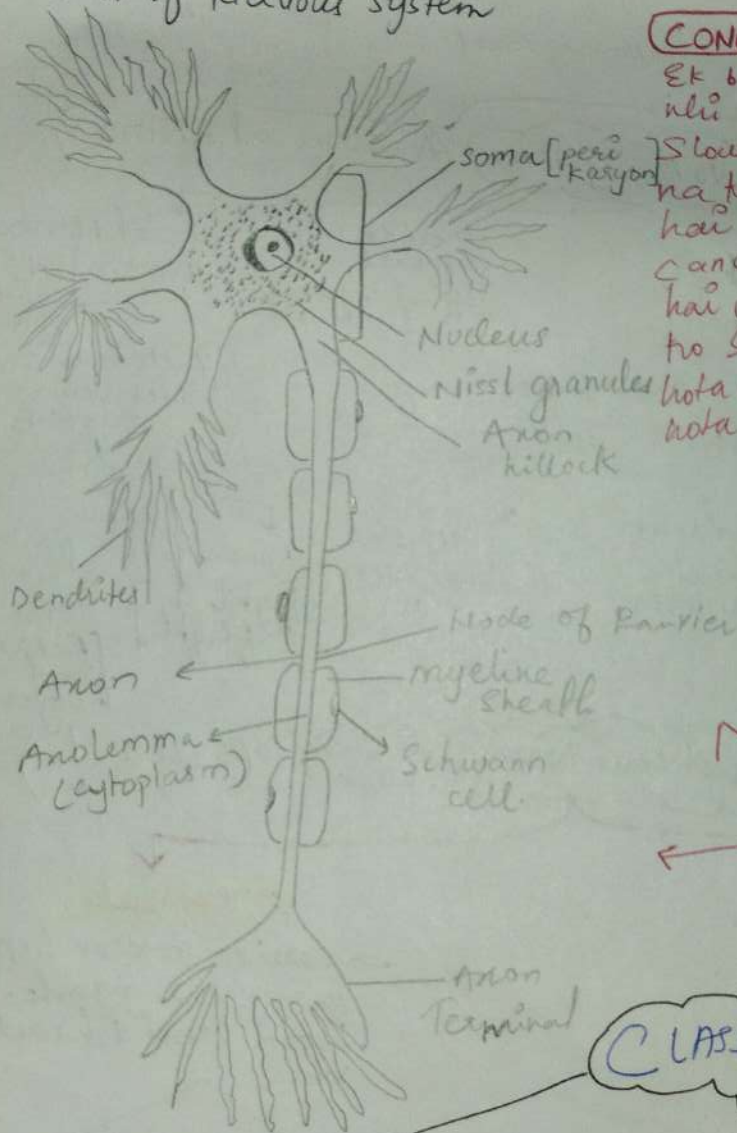
- Structural & functional Unit of Nervous System

Neuroglial cells or neuroglia

- Support the neurons.

CONCEPT :-

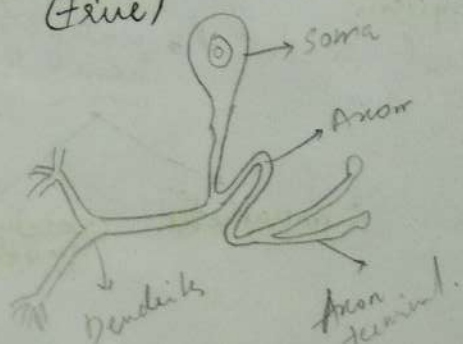
Ek baar nerve damage hogyi to Sahi nahi hoti kyunke neuron boht slow grow hote hai Ekdam na to barabar but hum bolte hai brain tumour hogya usmein cancer mein to excess growth hoti hai cells ki to wo kaise? To neuron ko support karne ke liye 1 cell hota hai neuroglial cell wo grow nahi hai [Glione] [oma-tumour]



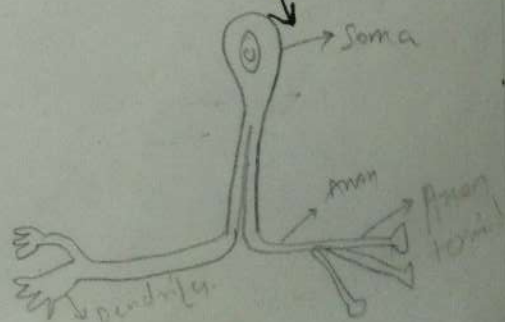
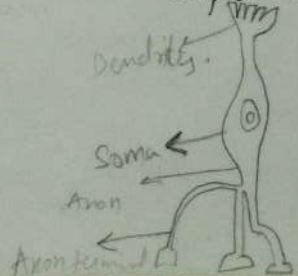
MULTIPOLAR NEURON

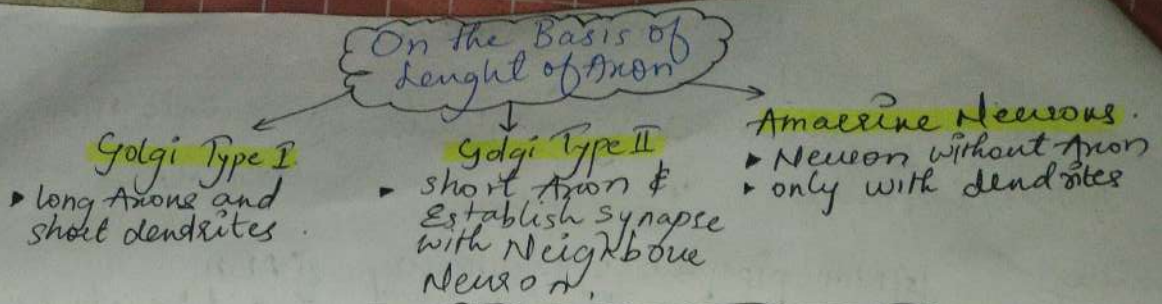
CLASSIFICATIONS OF NEURON (on the basis of no. of process)

Unipolar (true)

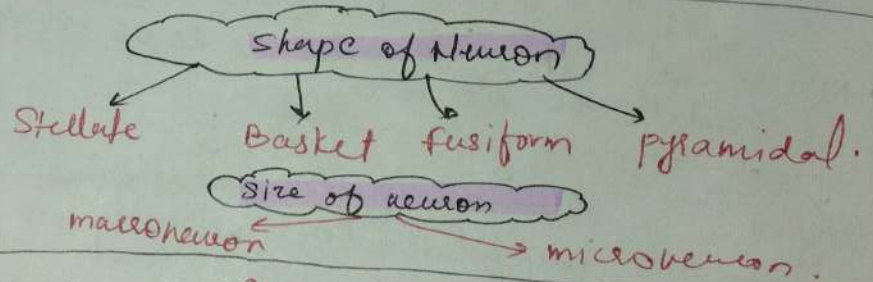
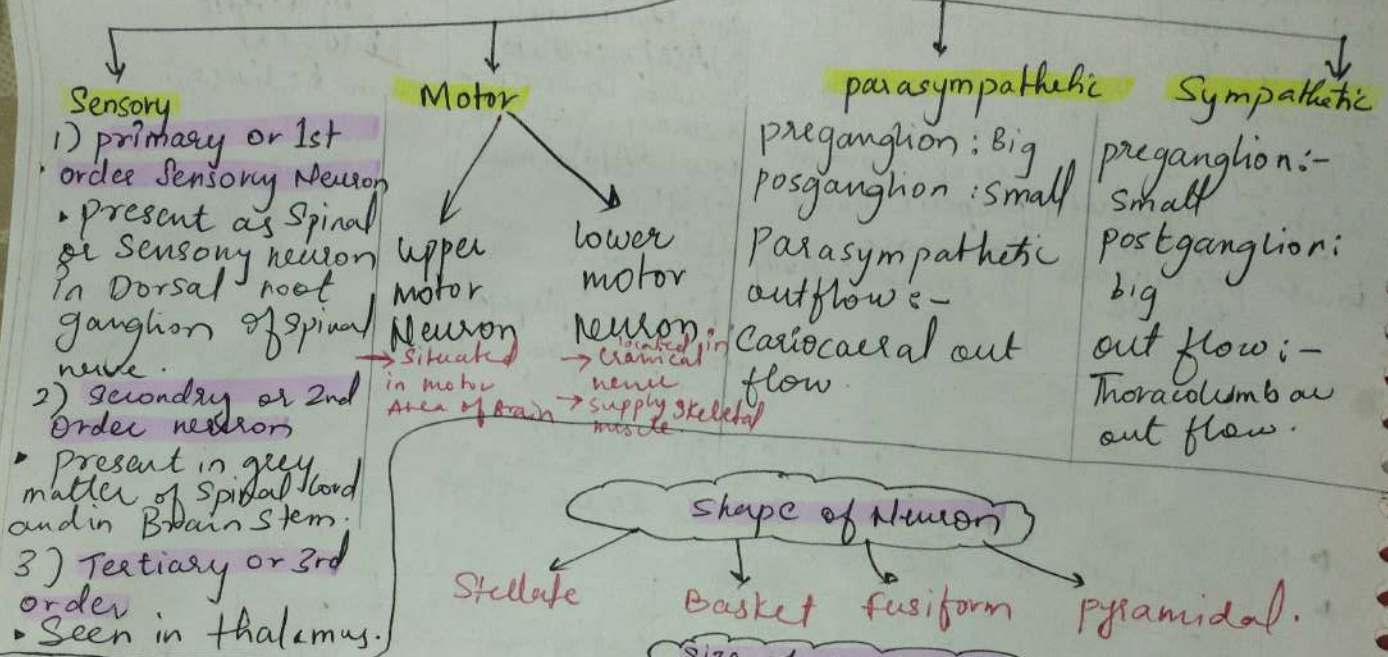


Bipolar

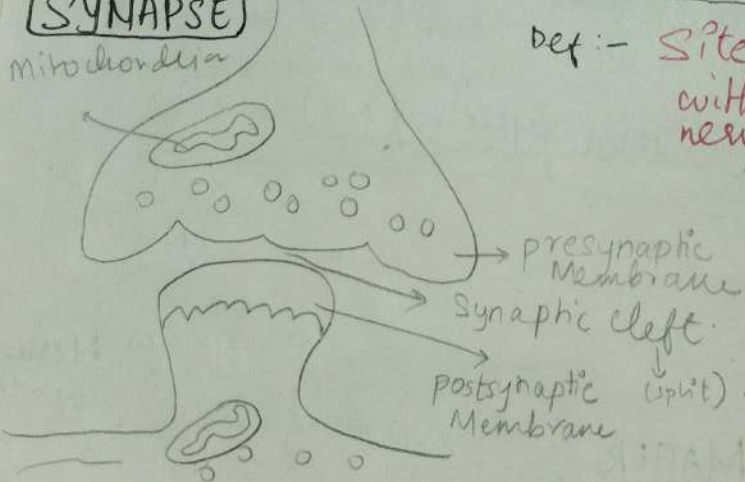




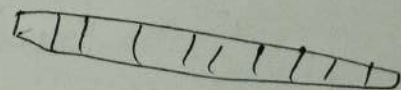
Functional Classification of Neurons



SYNAPSE

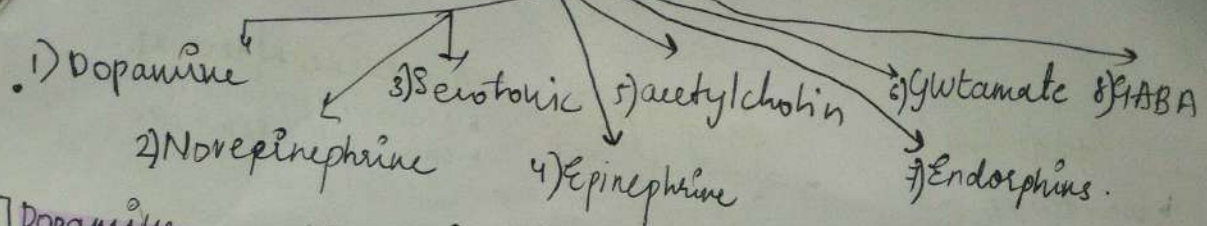


Def:- Site of Contact (contiguity without continuity) between nerve cell known as Synapse



Neurotransmitters:- Neurotransmitters are Chemical messengers that nerve cells / neurons releases to communicate with target cells such as another Neurons, muscle cells, glands across Synapse. They carry signals that regulate virtually Every function of Body & brain like movement, Emotion, heart Rate & learning.

Neurotransmitters



1] **Dopamine**
 plays imp role in Brain Reward System, motivation, compliments, having sex, complete task or few food.
 artificial :- Drugs, Social media likes etc.
 high :- impulsivity (ADHD)
 low :- parkinson's disease.

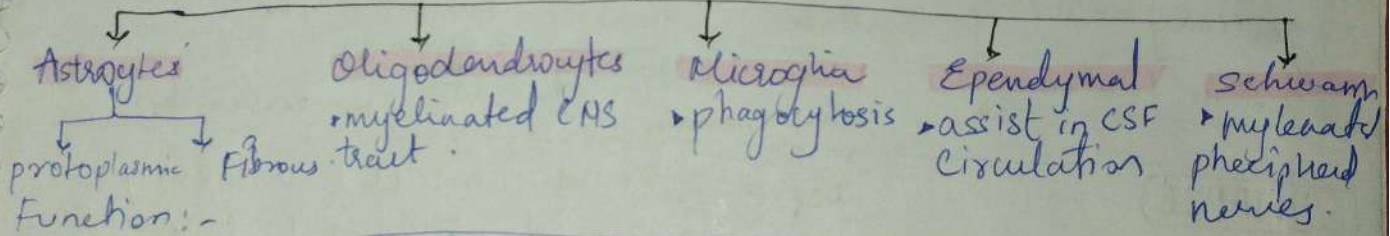
2] **Norepinephrine**
 Triggers bodies alertness, focus, fight or flight response.
 4] **Epinephrine**
 Similar to norepinephrine prepare body for more intense activity.

3] **Serotonin**
 Regulate mood, appetite, Digest & sleep.
 low :- depression, anxiety, insomnia.
 5] **Acetylcholine**
 muscle contraction, memory, learning.
 low :- Alzheimer's
 6] **GLU**
 Learning & memory.
 high :- neuron Damage
 high :- Epilepsy
 into release koga ke bits aajayegi

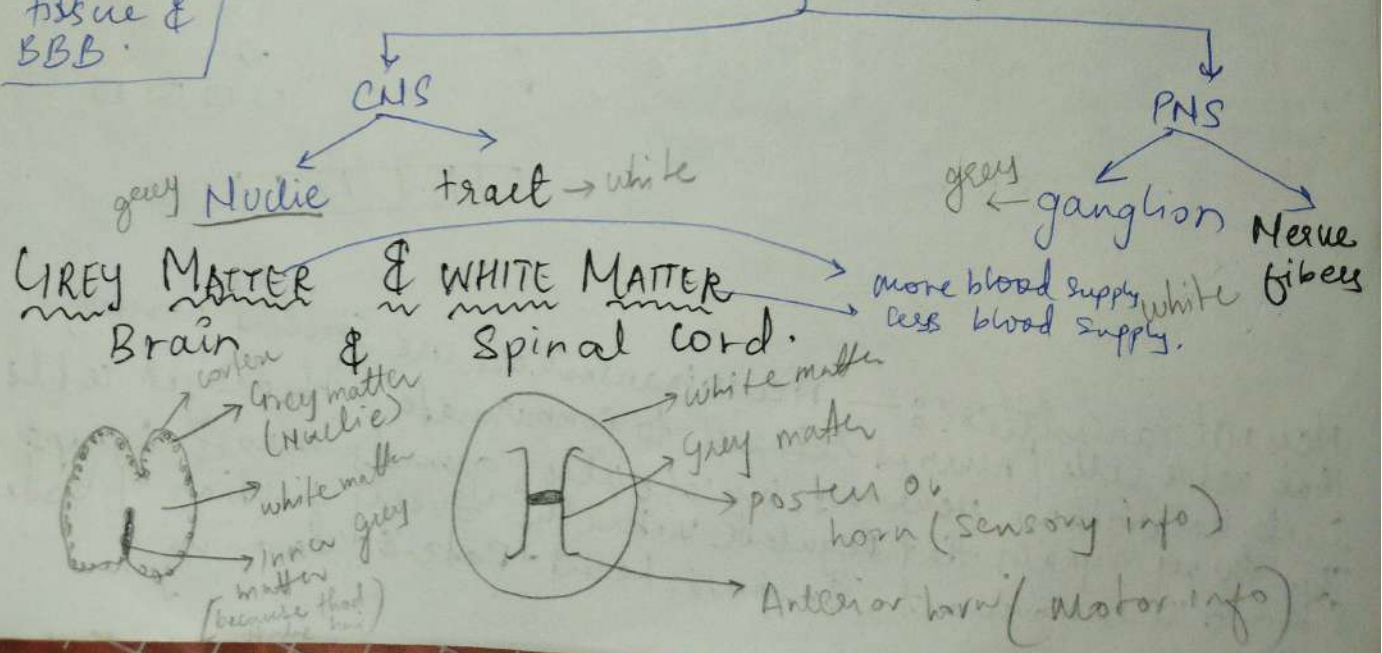
8] **GABA**
 Control Anxiety and Seizure.
 low level :- seizure, mood swing.
 7] **Endorphins**
 natural pain Relievers.

NEUROGLIAL CELLS

- Neuroglial cells also called as nerve glue
- it protect the Neurons

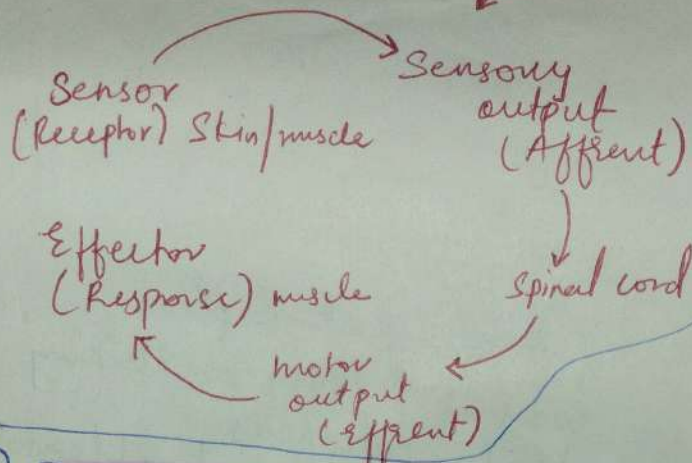


Nervous System (NS)



(REFLEX ARC)

- Brain does not involve in Reflex.
- Reflex are is functional unit of Nervous System.



REFLEXES

DEEP REFLEX

Deep reflexes are triggered by stretching of muscle tendon or muscle itself.
eg: biceps, patellar reflex.

SUPERFICIAL

on skin membrane
eg: abdominal, plantar, corneal Reflex.

SOME COMMON REFLEXES

1) Biceps Jerk reflex.

way to perform :- Striking biceps brachii tendon

Result :- flexion of Elbow joint.

if little flexion takes place it means its normal C5 & C6 nerve or brachial plexus Working properly

if Absent then there is Damage of C5-C6 roots, brachial plexus or biceps muscle itself.

if Exaggerated → UMN lesion.

Biceps Brachii Reflex



2) Triceps Jerk reflex.

way to perform : striking triceps brachii tendon.

Result :- Extension of Elbow joint.
if happens the C7, C8 is intact.



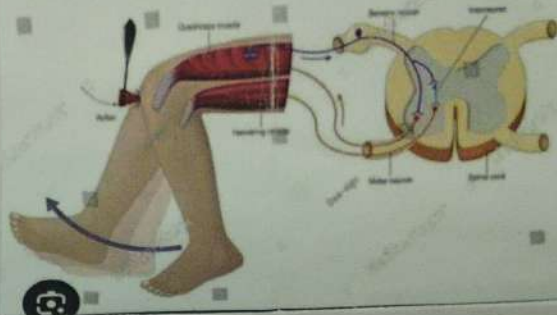
KNEE JERK REFLEX

way to perform:- striking the ligamentum patellae

Result:- Extension of knee joint.

its extension happens then L3, L4 is intact.

Knee-jerk reflex



Achilles Reflex

- Tests the S1, S2 nerve roots and the sciatic nerve
- Have your partner sit on a chair or table so that both feet hang over the edge
- Locate the achilles tendon of the right foot and tap it with the reflex hammer
- What happens?



4) Ankle Jerk Reflex

way to perform:- striking tendo calcaneus

Result:- plantar flexion of Ankle joint.
if it happens it means S1 & S2 Segment intact.

5) Abdominal Reflex

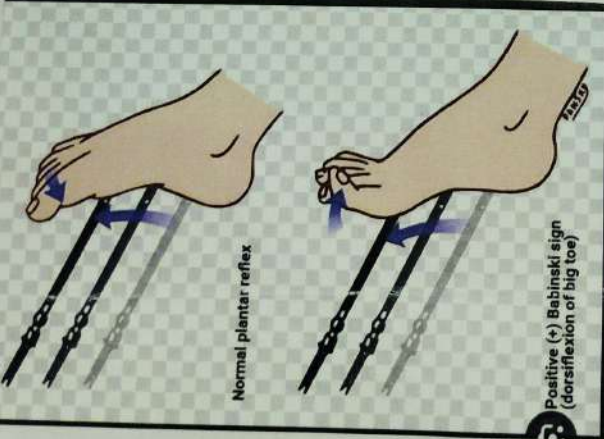
way to perform:- striking quadrants of Abdomen

Result:- contraction of Abdominal Muscle.

positive reflex indicates normal pyramidal tract with T7-T12 nerves intact.

Abdominal Reflexes

• Lightly & briskly stroke each side of the abdomen, above (T8, T9, T10) and below (T11, T12) the umbilicus.
• Note the contraction of the abdominal muscles and deviation of the umbilicus toward the stimulus.



6) plantar Reflex

way to perform:- scratching the sole of foot from lateral side towards big toe.
Result:- plantar flexion of great toe and other toe also.

This indicates (normal plantar response) intact pyramidal tract.

BABINSKI SIGN

Result:- Dorsiflexion of great toe and fanning of other toes.
this sign indicates pyramidal tract injury.

PARTS OF BRAIN (ENCEPHALON).

